

Egypt-Japan University of Science and Technology

School of Innovative Design Engineering/ Mechatronics & Robotics Eng. Dept.

MODULE ASSESSMENT

MODULE CODE	MTE 421
MODULE TITLE	Mechatronics Systems Design
MODULE TUTOR	Assoc. Prof.\ Mohamed G. Alkalla
SEMESTER/YEAR	2 nd semester 2025/2026
SUBMISSION DATE & TIME	21 st May 2026 – 18:00
VENUE/CAMPUS	Main E-JUST campus
SUBMISSION METHOD	Moodle Educational Platform
NOTES:	Any AI Assistance is PROHIBITED and will cause rejection of the submission.

Instructions to Candidates:

Value

This assignment constitutes 30% of the semester work grade for this module.

Submission Date and Time:

By 21st May 2026 – 18:00.

Submission Rules

The professional report generated in the assignment should be around **1500 words $\pm 10\%$** . The assignment should be your own work and written in your own words, and any information taken from other resources should be referenced. It will be checked for plagiarism using Turnitin. Any plagiarism or copying from others will be dealt with through the university's plagiarism procedures. **A prototype is required and is worth 10% of the total Mini-project points. The exceptional prototype will earn bonus marks.**

Your assignment must be submitted electronically via Moodle (Mini-Project tab) by the submission time/date. The software code will be submitted independently.

Any assignment submitted late but within 5 working days of the deadline will be given a maximum mark of 50%. Assignments submitted more than 5 working days after the deadline will not be marked, and a mark of 0% will be recorded.

Learning Outcome to be assessed:

1.	Identify and apply a range of engineering solutions to Mechatronics systems
2.	Critically analyse the practical aspects of sensors, actuators, and controllers
3.	Apply advanced-level knowledge and understanding of problems across a wide range of advanced Mechatronics systems
4.	Apply critical understanding of an appropriate and effective strategy for processing data, virtual instrumentation, and virtual simulation

Cooling and Sorting Automated Chocolate Bars Production Line

Problem Statement:

In modern food processing industries, particularly in chocolate manufacturing, products exit the moulding or forming stages at elevated temperatures and require controlled cooling to 20 °C to maintain quality, texture, and structural integrity. Improper cooling can lead to defects such as deformation, surface blooming, or compromised product consistency. Therefore, there is a need for an intelligent automated system capable of cooling, monitoring, and sorting chocolate products in real time.

This project aims to design and implement an integrated mechatronic system for an automated conveyor-based production line that transports chocolate bars, monitors their temperature, applies controlled cooling, and performs temperature-based sorting. The average and convenient linear speed of the conveyor should be 10 cm/s, and it should be adaptive to allow the chocolate bars to reach the desired temperature of 20 °C.

The system will convey chocolate bars along a belt where a temperature-sensing unit continuously measures their surface temperature. Based on this data, a cooling fan system will dynamically adjust its operation (on/off and variable speed control) to ensure that the chocolate bars reach an acceptable temperature range suitable for packaging. The inspection system should also include a measurement of the bar weight, which should be around 100 g. If not, a diversion mechanism should reject it and return it to the previous production stage, as shown in Fig. 1. After the cooling stage, a sorting mechanism will classify the products into:

Accepted products: Chocolate bars within the desired weight and cooling condition.

Rejected products: Overheated or insufficiently cooled bars and/or improper weight

Goal:

You are hired as a junior mechatronics engineer to develop a smart, energy-efficient, and reliable automated system that ensures chocolate bars are properly cooled and accurately sorted, improving product quality, reducing waste, and minimizing human intervention in industrial food processing

environments. The system must address several engineering challenges, including:

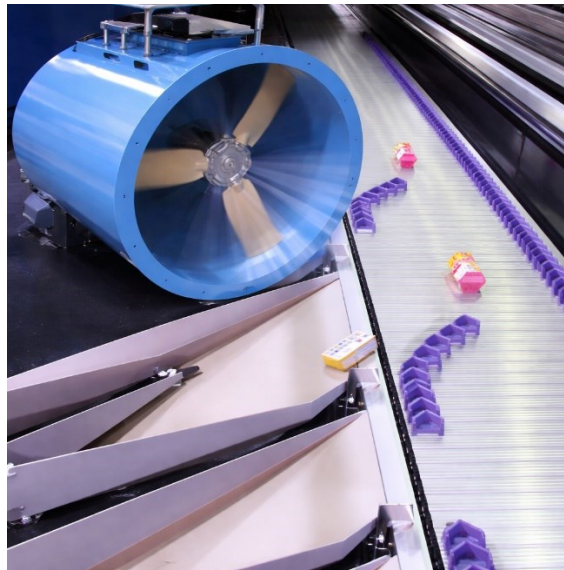


Fig. 1 Dummy automated line for chocolate bar sorting and cooling system

1. Conveyor speed control: Regulating belt speed to ensure sufficient cooling time and synchronization with sensing and sorting operations.
2. Temperature-based fan control: Designing a closed-loop control system where fan speed is adjusted based on real-time temperature feedback from chocolate bars.
3. Accurate temperature detection: Integrating reliable, non-contact sensors to measure the chocolate bar temperature without affecting the process.
4. Automated sorting mechanism: Designing a precise and responsive diversion system to separate acceptable and defective products.
5. Component selection: Proper selection and sizing of conveyor motors, fan motors, motor drivers, and control hardware based on load, speed, and performance requirements.
6. Mechanical design and CAD modelling: Developing a compact and efficient layout of the conveyor system, cooling unit, and sorting mechanism.
7. Force and load analysis: Evaluating mechanical stresses, torque requirements, and system stability. (At least 100 pieces are transported on the conveyor at any moment before transiting to the next packaging stage)
8. Mathematical modelling and simulation: Modelling conveyor dynamics, and simulating the system dynamically and control-wise.
9. Control system design and simulation: Implementing and validating control strategies (Preferred; PID controller) for conveyor and fan operation.
10. Prototype development: Building and testing a functional prototype to validate system performance under realistic operating conditions.

Main Objective

To design, simulate, and implement a fully integrated automated system that performs conveying, cooling, temperature monitoring, and intelligent sorting of products, as shown in the simplified and illustrative system in Fig. 2.

Specific objectives include:

- Design a controlled conveyor system with adjustable speed.
- Develop a temperature-based feedback system for real-time monitoring.
- Implement fan speed control based on product temperature.
- Design a sorting/diversion mechanism for classification.
- Select and size motors, drivers, and controllers appropriately.
- Develop mathematical models for system dynamics.
- Perform CAD design and mechanical analysis.
- Conduct dynamic and control simulations.
- Build and test a functional prototype.

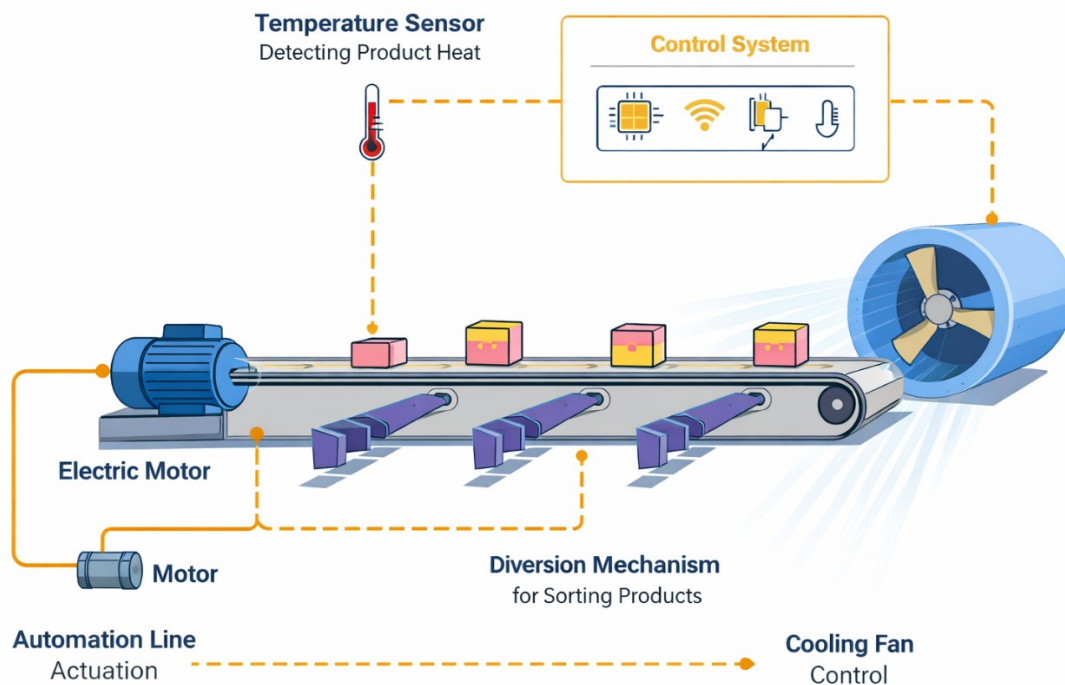


Fig. 2 Illustrative figure of the automated cooling and sorting chocolate bar production line

So, your mini-project work can be accomplished by achieving the following points as a guide to design an integrated mechatronics system:

1. Design a mechanism to convey the chocolate bars, as well as, the diversion or rejection mechanism (use free-hand sketch and CAD software). Select the suitable actuation system and mechanisms or kinematic chains.
2. Select the appropriate motor for the main rotation motion of the conveyor (*i.e.*, type, max. torque, rpm, etc.) with its gear head that will provide the required power to the system based on the calculations of force analysis. Demonstrate the detailed design & assembly of both the main rotation motion or conveyor elements (motor, bearing, coupling, conveyor, etc) and also the diversion mechanism's detailed design and its elements.
3. Select the most suitable sensors for monitoring the chocolate bar temperature and weight to be used as a feedback signal to the control unit.
4. Model the proposed mechatronics system (get the mathematical/dynamic model, define the inputs/outputs for your control system)
5. Using the Laplace Transform to model your system as block diagrams and all the subsystems, such as the conveying motion of the bars, the diversion system, and the cooling system.
6. Use multi-body dynamics simulation and control simulation tools for simulating the entire system (Preferably, Simulink/Simscape or ADAMS-Simulink co-simulation).
7. Validate your idea/design by using the simulation tools.
8. Propose the manufacturing processes and the practical implementation of your design.

For this task, you will need to address the following using referencing whenever required, as the following criteria will be used to evaluate the originality of your work:

1. Propose a solution to the above-mentioned system using the minimum required components, using the necessary mechanisms, Electronics, Computing, and Motor Drive to implement the mechatronic system in Figures 1 & 2.
2. Justify your choices for the required components by explaining the operational concept and why your choice fits the system requirements. Use all the necessary equations that describe your choice for:
 - Mechanisms
 - Electronics
 - Computing
 - Motor and Motor Drive

3. Provide detailed design and drawings with force analysis of the proposed system.
4. Provide the complete schematic diagram of the system and its corresponding representation in the Simulink or Simscape model.
5. Consolidate your system model using the transfer function reduction technique and simulate the resulting function.
6. Demonstrate the feasibility and practicability of your design over the traditional or manual sorting systems.
7. How to implement the procedures for fault detection of your system

Assessment Criteria

The Department's Principles of Assessment will be used to determine grading levels.

1.	Propose a solution to the above-mentioned system using the minimum required components, using the necessary mechanisms, Electronics, Computing, and Motor Drive to implement the mechatronic system in Figures 1 & 2. Explain your innovative idea.	10
2.	Justify your choices for the required components by explaining the operational concept and why your choice fits the system requirements. Use all the necessary equations that describe your choice for: - Mechanisms - Electronics - Computing - Motor and Motor Drive	10 10 10 10
3.	Provide detailed design and drawings with force analysis.	10
4.	Provide the complete schematic diagram of the system and its corresponding representation in Simulink or Simscape.	10
5.	Consolidate your system model using the transfer function reduction technique and simulate the resulting function.	10
6.	How to implement the procedures for fault detection of your system	10
7.	Demonstrate the feasibility and practicability of your design over the traditional or manual sorting systems. Implement a prototype of your work.	10

Best Wishes,